

Clustering Urban Mobility Patterns Using Bike-Sharing Data

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Abstract—Two years of hourly Capital Bikeshare records (17,379 observations, 2011–2012) are aggregated into 731 daily mobility vectors and clustered to identify recurring urban transport regimes. Nineteen candidate features are reduced to eleven by removing three algebraic identities and pruning collinearity until every variance inflation factor falls below five. K-Means, agglomerative clustering under three linkage criteria, and DBSCAN are compared on silhouette, Davies–Bouldin, intra- and inter-cluster distance, and size balance. The comparison shows that no single index is sufficient: the two most degenerate partitions obtained achieve the highest silhouette and Davies–Bouldin scores simultaneously. K-Means with $k = 5$ is selected on the combined criteria, yielding four substantive regimes—high-demand warm-season commuting, weekend leisure, cold-season commuting and adverse-weather suppressed demand—which recover the working/non-working distinction without being given any calendar attribute. Registered demand varies by a factor of 1.8 across regimes while casual demand varies by 4.9, and the regime is predictable one day ahead at 91.6 % accuracy from a weather forecast.

Keywords—clustering, urban mobility, bike sharing, K-Means, DBSCAN, hierarchical clustering

I. INTRODUCTION

Bike-sharing systems record urban movement in detail, but planning decisions are taken at the level of the day rather than the trip: how many bikes to deploy, when to reposition them, and when the network can be taken partly out of service. This study treats the day as the unit of analysis and asks what distinct mobility regimes exist in a two-year record.

The contribution is threefold: a feature-engineering pipeline in which every removal is justified by an algebraic identity or a measured collinearity; a comparison of three clustering families under a multi-criteria protocol shown to be necessary because individual indices reward degenerate solutions on this data; and validation of the selected clusters against calendar attributes withheld from the algorithms, translated into operational recommendations.

II. DATA PREPARATION AND FEATURE ENGINEERING

A. Cleaning

The hourly file contains 17,379 records over 731 consecutive days with no explicit missing values, no duplicate (date, hour) keys, and no out-of-domain values; the identity $\text{cnt} = \text{casual} + \text{registered}$ holds throughout. Two problems required action.

Of the 17,544 possible hourly slots, 165 were never written, and 59 % of these fall between 02:00 and 05:00—indicating that hours with no rentals produce no record rather than that

data was lost. The table was re-indexed onto the complete 731×24 grid with counts set to zero and weather left missing, so every day shares the same denominator for the proportion features. Eight days have eighteen or fewer recorded hours; each is a contiguous block coinciding with a documented weather emergency (the January 2011 snowstorms, Hurricane Irene, Hurricane Sandy). These are unusual but valid mobility days and were retained.

The only genuine data error is 2011-03-10, on which all 22 recorded hours report 0 % relative humidity—physically impossible, with neighbours at 78 % and 65 %. Readings were imputed from the same clock hour on adjacent days; plain time interpolation was rejected because both surviving endpoints are overnight humidity maxima, so interpolating returns a daily mean of 0.97, higher than either neighbour. No observation was deleted at any stage, and total rental volume is unchanged at 3,292,679.

B. Daily features

Nineteen features were constructed per day: the twelve required and seven additions. Peak windows follow the convention that a record labelled hour h covers $[h:00, h+1:00)$, so 06:00–09:00 is hours 6–8. The additions capture behaviour the required set cannot express. Night share (00:00–05:00) and midday share (11:00–15:00) disaggregate the off-peak block; peak-to-mean ratio and the normalised Shannon entropy of the 24-hour profile measure concentration independently of volume; apparent temperature, diurnal range and the share of adverse-weather hours refine the weather description. Weather features use physical units so that centroids remain directly interpretable.

C. Selection and scaling

Redundancy was removed in two stages. Three features are exact linear functions of others by construction: mean hourly rentals, off-peak share and registered proportion. These were removed unconditionally, since retaining them leaves the matrix rank-deficient. The remainder were pruned greedily on absolute correlation until every variance inflation factor fell below five, removing five more; a required feature was kept over an added one, and interpretability broke remaining ties.

Total rentals, standard deviation and maximum hourly rentals describe one busyness dimension, and the choice among them is near-arbitrary: the first two differ by only 0.001 in mean absolute correlation against the rest. Total rentals was kept for interpretability, and the choice verified immaterial, since the two candidate sets give pairwise-distance matrices correlating at 0.9987 over 266,815 day-pairs and four-cluster solutions agreeing at an adjusted Rand index of 0.954.

Skewness was assessed twice, over all days and excluding the eight storm days, to distinguish genuine asymmetry from a few extreme observations. Peak-to-mean ratio falls from +17.1 to +0.72 and entropy from -16.2 to -0.16 once eight of 731 days are set aside, so neither was transformed; a cube root was applied to the three features still skewed without them. Yeo-Johnson fits the skewness metric better but selects an exponent of -28.8 for night share, saturating the tail so severely that extreme days become indistinguishable from ordinary ones, and yields an identical final set. The eleven features (Table I) were then standardised, essential given ranges spanning four orders of magnitude.

TABLE I. The eleven clustering features (* = cube-root transformed).

Group	Features retained
Volume (1)	total rentals
Temporal shape (4)	morning peak share, evening peak share, night share*, hourly entropy
User mix (1)	casual proportion*
Weather (5)	mean temperature, humidity, wind speed, temperature range, adverse-weather share*

III. EXPLORATORY FINDINGS

Principal component analysis on the standardised matrix explains only 47.6 % of variance in two dimensions and requires seven components for 90 %, so projections were treated as suggestive rather than as evidence. PC1 (25.2 %) is a commuting-leisure axis, correlating +0.885 with casual proportion and -0.743 with morning peak share but only +0.381 with total rentals. PC2 (22.4 %) is a demand-and-conditions axis, correlating +0.664 with total rentals and loading negatively on adverse weather. That these are separate axes is itself a finding: a day's character and its demand level vary largely independently.

Three further observations shaped the analysis. Working days show a sharply bimodal profile with peaks at 08:00 and 17:00–18:00 while non-working days show no morning peak and a single 11:00–16:00 plateau. Demand grew 64 % between 2011 (3,406 rentals/day) and 2012 (5,600), a trend not modelled because days are treated as exchangeable. Most importantly, the known working/non-working partition—which genuinely exists—achieves a silhouette of only 0.237 in the full space. This provides a calibration point: modest silhouette values are the realistic expectation on this data, and a markedly higher score warrants scrutiny rather than confidence.

IV. CLUSTERING METHODS AND PARAMETER SELECTION

A. K-Means

Five criteria were applied over $k = 2 \dots 10$ and they disagree. Silhouette and Calinski-Harabasz both favour $k = 2$; the normalised knee of the inertia curve indicates $k = 6$, with $k = 5$ tied to within 0.0003; Davies-Bouldin is minimised at $k = 5$. Two structural facts resolved the disagreement. First, $k = 2$ is a false optimum—it merely reproduces the working/non-working split already measured at silhouette 0.237, so its score of 0.252 represents no discovery. Second, from $k = 4$ onward K-Means allocates an entire cluster to Hurricane Sandy, at

every k tested; this is a property of an algorithm that must assign every point, not an artefact of the chosen k . $k = 5$ was selected, giving four substantive clusters (247, 208, 150, 125 days) and the single-day cluster.

B. Agglomerative clustering (AGNES)

Single, complete and average linkage were applied with Euclidean distance and cut at $k = 2 \dots 10$. Only complete linkage produces a usable partition at any cut. Single linkage never escapes the chaining effect: at $k = 10$ the largest cluster still holds 722 of 731 days and the nine isolated points are mostly storm days, so it functions as an outlier detector rather than a clustering algorithm. Average linkage behaves similarly but more slowly, holding 91.4 % of days in one cluster at $k = 7$. Complete linkage penalises cluster diameter and is therefore the only criterion forced to divide the bulk.

Cutting at $k = 5$ and $k = 6$ is metrically indistinguishable (Davies-Bouldin 1.765 against 1.761; Calinski-Harabasz 109.9 against 110.9). The tie was broken by external agreement: adjusted Rand index against the K-Means solution rises from 0.276 to 0.436 at $k = 6$. Convergence between two algorithms with different objectives is evidence about the data, and is not obtainable from dendrogram inspection.

C. DBSCAN

With $d = 11$ features, min_samples was set to $2d = 22$, equal to 3 % of the record. The knee of the 21st-nearest-neighbour distance curve, located numerically, gives $\text{eps} = 3.57$. At these parameters DBSCAN returns one cluster and eight noise points. Across 78 settings the output is governed almost entirely by eps, and there is no stable two-cluster regime with acceptable noise: forcing a second cluster requires $\text{eps} = 2.0$, which discards 21.2 % of days. The commuting and leisure groups are therefore separated by a low-density region rather than an empty one—enough intermediate days exist to keep them density-connected.

V. EVALUATION AND COMPARISON

DBSCAN is scored on its clustered points only, with noise excluded and the excluded count reported. The alternative convention—treating noise as a cluster—raises the knee setting's silhouette to 0.588, the highest value obtained anywhere in this study, for a result containing one real cluster. The convention must therefore be declared rather than assumed. It also means DBSCAN is not judged on the same task, having discarded its hardest cases before measurement.

Table II reports the required metrics. Ranked by silhouette alone the two worst clusterings obtained occupy first and third place, and AGNES single linkage leads on silhouette and Davies-Bouldin simultaneously while placing 725 of 731 days in one undifferentiated group. Distance structure does not detect this either, since the gap from a large blob to an isolated point genuinely is large relative to the blob's spread. Only size distribution is decisive, with normalised entropies of 0.034 and 0.064 against 0.845 for K-Means. Four criteria were required to reach a conclusion any one alone would have got wrong.

TABLE II. Evaluation. Gap = closest centroid separation; Ent. = normalised size entropy. DBSCAN scored on clustered points only.

Result	k	Sil.	DB	Intra	Gap	Ent.
K-Means	5	0.199	1.280	3.26	2.62	0.845
AGNES complete	6	0.147	1.263	3.52	2.64	0.646
AGNES single	6	0.312	0.411	4.20	4.74	0.034
AGNES average	6	0.274	0.750	4.17	5.18	0.064
DBSCAN eps=2.0	2	0.293	1.416	3.22	3.24	0.845
DBSCAN eps=3.57	1	—	—	—	—	—

On the six required comparison criteria, K-Means is best on compactness (intra-cluster distance 3.26 against a dataset mean of 4.34), size balance, and parameter stability (adjusted Rand 0.896 under 80 % resampling, against 0.334 for AGNES complete and a collapse to 0.266 for DBSCAN when eps rises 10 %). Its separation is comparable to AGNES complete. It is clearly worst on treatment of unusual observations, absorbing seven of the eight storm days and spending a cluster on the eighth, where DBSCAN correctly labels all eight as noise. Interpretability favours K-Means, which yields four operationally distinct regimes against DBSCAN's two.

K-Means with $k = 5$ was selected. It does not lead on any single index and is not chosen as though it did. AGNES single and average are excluded on size distribution; AGNES complete is a close second recovering similar structure but is rejected on reproducibility, since a resampling agreement of 0.334 means a different sample would yield a substantially different partition; DBSCAN is rejected because two clusters cannot support demand-tier planning and its solution is unstable. The known weakness is carried forward explicitly: Part VI treats the single-day cluster as a flagged anomaly and uses DBSCAN's noise set as the authority on exceptional days. The two methods are complementary—K-Means supplies the partition, DBSCAN the exception list.

VI. INTERPRETATION OF THE SELECTED CLUSTERS

Table III profiles the four substantive clusters in original units. None of the calendar or weather attributes used for validation was available to the algorithm.

TABLE III. Cluster profiles in original, non-standardised units.

	C0	C1	C2	C3
Days	247	208	150	125
Rentals/day	5,867	4,516	3,449	3,095
Busiest hour	691	425	423	405
Morning peak share	0.181	0.036	0.187	0.185
Midday share	0.204	0.421	0.219	0.219
Casual proportion	0.157	0.310	0.085	0.099
Temperature (°C)	21.8	15.5	6.6	12.4
Humidity (%)	61.7	61.8	53.2	78.3
Adverse-weather share	0.027	0.053	0.021	0.315
Working days (%)	98.4	1.0	94.7	89.6
Weekend days (%)	0.4	93.8	2.0	8.8

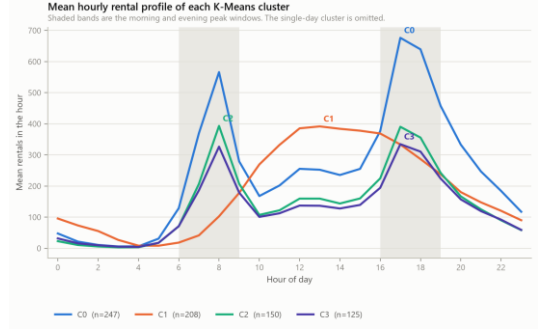


Fig. 1. Mean hourly rental profile of each cluster. Shaded bands are the peak windows. C1 has no morning peak; C0, C2 and C3 share one shape at descending amplitude.

The clusters are named high-demand warm-season commuting (C0), weekend and holiday leisure (C1), cold-season clear-weather commuting (C2) and adverse-weather suppressed demand (C3). External validation is strong: C1 contains 92.4 % of all Saturdays and 93.3 % of all Sundays while being 99 % non-working, and the three commuting clusters are 89–98 % working days. C0 is 47 % summer and 3 % winter; C2 is 57 % winter and contains no summer day at all; C3 holds 30 of the dataset's 33 light-precipitation days. The working week was reconstructed from rental timing alone.

One result is not a confirmation. Of 21 public holidays, eleven fall in the leisure cluster but ten fall in commuting clusters, so the holiday flag alone is a poor predictor of mobility behaviour. This is discoverable only because the calendar was withheld.

Behaviourally, C1 is the most distinctive: its morning window carries 163 rentals against C0's 1,064, a 6.5-fold difference, yet its daily total of 4,516 is second highest. Leisure days are busy at different hours and served by a different population, with casual riders supplying 1,480 rentals per day. C2 is behaviourally identical to C0 at 59 % of the volume, and the difference lies in who disappears: registered demand falls 36 % while casual demand falls 67 %. Across all four regimes registered demand varies by a factor of 1.8 and casual demand by 4.9, so essentially all variance in daily demand is casual-rider variance.

VII. PRACTICAL MOBILITY INSIGHTS

1) Size the fleet for one regime, not the average. The busiest hour averages 691 rentals in C0 and reaches 977, against 405–425 elsewhere; at the 90th percentile the gap is 890 against 656–680. Provisioning uniformly for C0 leaves roughly 222 bikes per hour idle on the other 483 days. A reserve tranche deployed only on forecast-C0 days converts that idle capacity into maintenance throughput without reducing peak service.

2) Rebalancing should switch mode, not merely scale. The evening-to-morning peak ratio is 1.59, 1.45 and 1.44 for the commuting regimes but 6.07 for leisure days, where the morning window carries only 163 rentals. Commuting days present a directional problem—reversing morning accumulation before the larger evening peak—whereas leisure days require continuous low-intensity redistribution through 11:00–16:00. These are different operations.

3) Maintenance windows are predictable. Ninety-seven days fall below 2,000 rentals—36 in C2, 34 in C3, 27 in C1, none in C0—amounting to 13.3 % of the year. C2 concentrates in January, February and December, but C3 occurs in all four seasons, so an adverse-weather day in June is also a legitimate slot that a season-based rule would miss.

4) The regime is forecastable a day ahead. A random forest predicts cluster membership at 91.6 % accuracy (five-fold cross-validated, 33.8 % majority baseline) from a weather forecast plus the calendar, with working-day status, temperature and adverse-weather share the dominant predictors. This is what makes the preceding three recommendations actionable: a four-way classification is a far more robust forecasting target than a continuous rental count.

These insights concern system-wide daily demand. The dataset contains no station-level data, so recommendation 2 identifies the type and timing of the required operation but not which stations to service.

VIII. THEORETICAL DISCUSSION

A. K-Means assumptions. Assignment to the nearest centroid makes boundaries perpendicular bisecting hyperplanes, so clusters are convex and implicitly isotropic; elongated or nested shapes cannot be represented. Minimising within-cluster sum of squares penalises large diameters, biasing towards clusters of comparable extent. The mean minimises squared Euclidean distance, so the method is coherent only under L2 and only when features are comparably scaled.

B. Effect of the linkage criterion. Linkage defines cluster-to-cluster distance and hence the merge order and the whole tree; since merges are greedy and irreversible, the bias compounds. It encodes what counts as a cluster: connectedness (single), small diameter (complete) or aggregate closeness (average). Identical data and metric here gave a 722-day chain, a 91 % blob and a usable 44 % partition.

C. Single-linkage chaining. Cluster distance is the minimum over cross-cluster pairs, so one intermediate point merges two groups however far apart their bulks lie, and merge cost never grows with cluster size. Equivalently, single linkage cuts the minimum spanning tree, so any path of short edges connects distant regions. On a continuous cloud only points with no near neighbour can be separated.

D. Why DBSCAN identifies noise. K-Means optimises over partitions, where every point belongs to exactly one cluster by definition; the objective has no reject option, so a distant point joins the nearest centroid and distorts it. DBSCAN defines membership by a local density predicate, so a point failing it and not density-reachable from a core point belongs to no cluster. Noise is definitional.

E. Necessity of feature scaling. Retained features span four orders of magnitude, from 22-8,714 daily rentals to 0-0.381 morning peak share. Euclidean distance sums squared differences, so contribution scales with squared range and the daily total would otherwise supply almost the entire distance. All three algorithms are distance-based, and PCA would likewise return total rentals as PC1.

F. Effect of correlated features. Correlated features measure one dimension repeatedly, so it enters the distance sum several

times and is silently up-weighted, undoing the equal weighting standardisation was meant to establish. Four volume features correlating 0.93-1.00 would have given busyness four times the weight of humidity, and three exact identities made the matrix rank-deficient.

G. PCA for visualisation only. Advantages: no variance is discarded, which matters because two components capture only 47.6 % here; discriminative structure need not be high-variance, since PCA maximises variance rather than separability; and centroids stay interpretable in physical units. Disadvantages: reduction can mitigate distance concentration, suppress noise dimensions and decorrelate features automatically. The risk is over-reading a projection, as occurred here when a t-SNE island suggested a density gap absent in eleven dimensions.

H. Manhattan versus Euclidean distance. L1 sums absolute rather than squared differences, so one large coordinate deviation is penalised far less and results are more robust to outliers. Unit balls become cross-polytopes, giving more axis-aligned clusters, and since the mean minimises only L2, L1 requires k-medians or k-medoids. Hurricane Sandy lies 22 standard deviations from the mean on entropy; under L1 that contributes linearly, so it would probably not have formed its own cluster.

I. Why not a single metric. Each index encodes one definition of quality and can be maximised by useless solutions. At $k = 2$ all three linkages scored silhouette 0.831 and Davies-Bouldin 0.411 for a partition of 730 days plus one; single linkage led both indices while placing 725 days in one group; cophenetic correlation ranked the linkages in reverse order of usefulness; and degenerate solutions even scored highest on resampling stability. Only size distribution exposed this.

J. Unusual valid days versus data-quality outliers. Both are extreme in feature space, so magnitude cannot separate them; the distinction is provenance. A data-quality outlier means the measurement process failed, so it belongs to the cleaning stage: 2011-03-10's impossible 0 % humidity was repaired. An unusual valid day is a correct measurement of a rare event and must be retained: 2012-04-22 has a flawless 24-hour record yet only 1,027 rentals under all-day rain. Decisively, the repaired error was not flagged as noise by DBSCAN, whereas every day it did flag was genuine signal.

IX. REPRODUCIBILITY AND CONCLUSION

All results were produced by a single Jupyter notebook that executes from beginning to end with `RANDOM_STATE = 42` fixed for K-Means initialisation, t-SNE, PCA, random-forest validation and all resampling procedures. Software: Python 3.13.7, NumPy 2.4.6, pandas 3.0.3, scikit-learn 1.9.0, SciPy 1.18.0, Matplotlib 3.11.0, seaborn 0.13.2.

Cleaning and transformation decisions are documented inline with justifications: the zero-fill of 165 unrecorded hours, same-hour imputation of the 2011-03-10 sensor fault, retention of all eight storm days, removal of eight features with the identity or correlation underlying each, cube-root transformation of three features, and z-score standardisation. No observation was removed at any point, and none to improve a clustering score: cleaning added 165 explicit zero rows and repaired 22 readings, leaving all 731 days and all 17,379 original observations in the analysis.

Four recurring mobility regimes were identified from behavioural and weather features alone and validated against calendar attributes withheld from the algorithms. Methodologically, the dataset admits no strong density structure—DBSCAN returns one cluster at its knee—so the regimes are partitions of a continuum rather than natural kinds, and evaluation must be multi-criteria because individual indices reward degenerate partitions. Practically, demand variance is concentrated almost entirely in casual riders, and the operating regime is predictable a day ahead at 91.6 % accuracy, making regime-conditional planning feasible.

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